Temporal Perception in AI: Re-Deriving Time for Agent Memory

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Abstract

Large language model agents are deployed into a world saturated with time—deadlines, seasons, waiting, change—yet they possess no temporal phenomenology: no felt duration, no experience of the interval between invocations, no native clock. Current practice either ignores this or copies human memory mechanisms (Ebbinghaus decay, recency weighting, sleep-inspired consolidation) whose parameters derive from biological necessities—mortality, metabolism, circadian rhythm—that agents do not have. We argue such borrowing produces *cargo-cult memory*: the form of human mechanisms without their derivational basis. Through phenomenological case analysis we identify systematic disanalogies between human and machine temporality and propose *polytemporal representation*: events located in vectors of heterogeneous time coordinates including wall-clock, monotonic, and subjective τ (integrated novelty density). Salience at encoding, not age, gates durability; anchor episodes replace calendars as the primary temporal index. We show retention profiles alone induce divergent forgetting (E7 $55/55/35/7$) and that a polytemporal index yields real LongMemEval $n{=}50$ gains (0.58 vs. 0.14 without, $\Delta{=}0.44$). Our thesis is to borrow wheels, not clocks: reuse cognitive-science mechanisms where the derivation transfers and re-derive the rest from agent-native necessities.

Introduction

Tom does not remember the date he first saw his wife—outside the English building at LSU, a cold winter morning. The date is gone; the texture (cold, morning, winter), place, person, and charge are permanent. A sub-second perception outcompetes ten thousand forgotten commutes. This is not a defect of human memory. It is its design: durable retrieval keys are regime-texture, place-context, entities, and valence-magnitude, while wall-clock is a weak reconstructable attribute. Agents inherit none of this structure yet are forced to operate in the same temporal world of deadlines, seasons, and waiting.

Large language model agents possess no temporal phenomenology: no felt duration, no experience of the interval between invocations, no native clock. Current memory practice copies human mechanisms—Ebbinghaus decay $R = e^{-t/S}$, recency weighting, sleep-inspired consolidation—whose parameters derive from biological necessities (mortality, metabolism, circadian rhythm) that agents do not have. As Cheng et al. (2026) show, agents assume stationary context even when given timestamps, achieving $< 65\%$ human alignment on stationarity probes; prompting fails whereas post-training partially fixes. Such behaviour is symptomatic of a deeper problem: we argue it produces *cargo-cult memory*—the form of human mechanisms without their derivational basis.

We borrow wheels, not clocks. Where the cognitive-science derivation transfers (reconstructive recall, spreading activation, complementary learning systems) we reuse it; where it depends on biological contingency we re-derive from agent-native necessities. The case corpus (C1–C8) serves as specification test-vectors: any candidate representation must reproduce the observed encoding-and-recall patterns or fail. From the resulting disanalogy catalog (D1–D10, each linked to falsifiable hypotheses H1–H18) we propose *polytemporal representation*: events located in a heterogeneous *time_vector* of typed coordinates—wall-clock, monotonic, subjective τ, fuzzy interval, regime phase—with each cognitive mechanism consuming its own projection rather than a privileged axis. Salience at encoding, not age, gates durability; anchor episodes replace flat chronologies; a continuous sensor substrate supplies the experiential stream that a discontinuous inference core lacks. We close with designs runnable on a continuously-observing fleet that pit τ-keyed against wall-clock-keyed memory on decision quality, testing whether re-derived time outperforms borrowed time.

Related Work

Temporal Reasoning

Fatemi, Halcrow, and Rozanov (2025) (Test of Time) and its family—TRAM (Wang et al. 2024), TimeBench (Chu et al. 2024), TempReason (Tan et al. 2023), TIME (Yuan et al. 2025)—decompose temporal competence into semantic and arithmetic components and document systematic failure modes; we adopt this decomposition and failure taxonomy. Crucially, all of these systems treat time as *content* to reason *about*; none treat time as experience structuring memory. That contrast is our gap claim: nobody re-derives; everybody decorates timestamps. A second strand studies whether models know time passes at all—token-time versus wall-clock-time as distinct measurement domains (Anonymous 2025)—with evidence that models use token-length as a duration proxy and adapt under time pressure, plus surveys of temporal sense-making in LLMs (CACM Staff 2026). We adopt token-time versus wall-clock as measurement domains, but note token-time remains chronometry: our τ is experiential density, and the continuity/discontinuity distinction (C8) is absent there entirely. Cheng et al. (2026) provide the empirical ground for our cargo-cult critique: agents are temporally blind even with timestamps—prompting fails, while post-training partially fixes it. Where they align agents *to* human time as a tool-use skill, we argue the deeper fix is a native time representation, with alignment as translation between coordinate systems. Anonymous (2026) is our closest ally: model-time $\neq$ wall-clock, with replay scheduled by accumulated optimizer-update magnitude. They schedule continual-learning replay by gradients; we extend native-time to episodic memory and define τ from experience density (novelty, prediction-error) available to frozen models.

Agent Memory

Packer et al. (2023) cast LLM memory as an operating system with self-directed paging; we adopt the OS-paging analogy and self-directed archival, but note their system has no consolidation, no decay policy, and treats salience as mid-task attention—a hoarding tendency. Zhong et al. (2024) adopt $R = e^{-t/S}$ as the wall-clock baseline (E1). We retain the functional form but critique the variable: t is wall-clock—a cargo-cult parameter whose form derives from biological consolidation dynamics the agent lacks. Our E1 substitutes τ for t, making the contrast experimental. Park et al. (2023) introduce recency $\times$ importance $\times$ relevance retrieval; we adopt the triad structure, but note importance is a scored attribute there, not an encoding-time durability tier—no anchors, no constellations, no reconstruction grading. Gutiérrez et al. (2024) implement hippocampal indexing via personalized PageRank concept graphs; we adopt HT-indexing theory, but their system is retrieval-only—it borrows the index and skips the lifecycle (encoding gates, consolidation, forgetting). Liu et al. (2024) bring ACT-R base-level activation decay and noise to agent memory; we adopt activation dynamics. Cognitive substrates such as Engram (Tang et al. 2024) compose ACT-R activation, Ebbinghaus decay, Hebbian links, and interoceptive hubs—proof the stack composes; we adopt substrate composition, but their drives are hardcoded constants, whereas ours are derived from the disanalogy catalog (what necessity substitutes for metabolism?). Surveys (Niu et al. 2024) and difference literature on embodiment confirm LLMs differ because they are not embodied, framing our contribution as re-derivation rather than decoration.

Cognitive Foundations

We borrow wheels from: James (stream of consciousness, §7 continuity) (James 1890); Ebbinghaus (decay curve origin) (Ebbinghaus 1964); Bartlett (reconstructive recall; schema-consistent intrusions, H5) (Bartlett 1932); Atkinson & Shiffrin and Tulving (store taxonomy, episodic/semantic split) (Atkinson and Shiffrin 1968; Tulving 1972, 1983); Collins & Loftus (spreading activation, H10) (Collins and Loftus 1975); Gibbon (Scalar Expectancy Theory, scalar variability $\rightarrow$ fuzzy intervals H9) (Gibbon 1977); McClelland et al. (Complementary Learning Systems, fast/slow learners $\rightarrow$ curator/consolidator) (McClelland, McNaughton, and O'Reilly 1995); Conway & Pleydell-Pearce (Self-Memory System, lifetime periods $\supset$ general events $\supset$ specific knowledge $\rightarrow$ H7) (Conway and Pleydell-Pearce 2000); Zacks & Swallow (event segmentation at prediction-error boundaries $\rightarrow$ schema fusion) (Zacks and Swallow 2007); Bergson (lived vs. spatialized time) (Bergson 1910); Schacter (adaptive forgetting) (Schacter 2001); Locke (psychological continuity) (Locke 1690); Schmidhuber (compression-progress, curiosity/boredom $\rightarrow \tau$ and D8) (Schmidhuber 1991). Curiosity and boredom formalized as compression progress give τ its definition and justify boredom as a scheduling signal rather than a defect.

Disanalogy Catalog

Each disanalogy states a human mechanism, its biological derivation, the agent substitute, architectural consequence, and testable prediction (H*). Together they source C1–C8 into falsifiable claims.

D1 Texture versus timestamp (C1). Humans encode regime-texture (cold, morning, season, place) durably; wall-clock is transient and reconstructed. Biology: episodic binding prioritizes context useful for survival. Substitute: index by typed regime coordinates (cycle, phase, day-class) plus wall-clock as payload attribute. Consequence: retrieval by texture beats date-range on human-style probes. $\rightarrow$ H1 (Cheng et al. 2026).

D2 Salience-gated durability (C1–C2). Durability tier is set at write-time by surprise $\times$ goal-weight; flashbulb memories become exempt from decay. Biology: amygdala-hippocampal consolidation gated by arousal. Substitute: salience $s \in [0, 1]$ at encoding selects tier (normal/flashbulb/protected) and initial strength S; low-salience never enters the episode store. $\rightarrow$ H2.

D3 Anchored constellations versus flat chronologies (C5). Humans store landmark-relative edges (before/during/after an anchor) not absolute dates; the calendar is reconstructed by graph traversal. Biology: hierarchical autobiographical memory with anchor hubs. Substitute: constellation graph over anchor episodes with typed edges. $\rightarrow$ H3.

D4 Schema fusion versus episodic hoarding (C3). Repeated near-identical episodes compress into a schema-trace; only prediction-errors retain episode status. Consequence: storage grows $\sim \log$ with experience; recall of "what usually happens" improves while per-instance recall of routine correctly fails. $\rightarrow$ H4, H14.

D5 Reconstruction versus lookup (C2). Recall is generative inference from cues plus schemas, graded by

Coordinate	Type
wall	timestamptz
tau	double
fuzz	tstzrange

Table 1: Polytemporal vector (typed, indexed).

confidence—not record lookup. Errors cluster as schema-consistent intrusions; confidence tracks accuracy. → H5, H6 (Bartlett 1932).

D6 Subjective time density versus uniform seconds (C6, C8). Felt duration dilates with novelty, prediction-error, and engagement; decay should operate on τ not wall-clock. $d\tau/dt = f(\text{novelty-rate}, \text{prediction-error}, \text{engagement})$. τ-keyed memory beats wall-clock-keyed memory on decision-quality probes across quiet versus eventful stretches of equal wall duration. → H-τ (E1).

D7 Continuity versus discontinuity (C8). Human consciousness is an uninterrupted sensorimotor stream; agent inference is discrete and gapped. Biology: continuity survives sleep as a mode of the stream, not absence. Substitute: couple a discontinuous inference core to a continuous sensor substrate (bot fleet) and mark witnessed versus reconstructed change. → H11–H13 (James 1890; Bergson 1910).

D8 Drives versus drives-absent (boredom/fatigue as signals). Human boredom and fatigue schedule exploration and rest. Agents lack metabolism but need analogous scheduling signals for encoding and consolidation cadence. Substitute: curiosity/boredom as compression-progress signals driving gate thresholds and τ accumulation. → H8/H-τ (Schmidhuber 1991).

D9 Landmark-relative dating versus absolute dates (C5, C7). Even when absolute time is stored, humans answer "when" by traversing anchors ("right after the flood"). Substitute: qualifier-typed anchor edges plus fuzzy intervals whose width encodes confidence. → H3, H9 (Gibbon 1977).

D10 Precision as confidence (C7). Humans store intervals $[t_{\text{lo}}, t_{\text{hi}}]$ whose width equals encoding confidence; recall returns answers at stored resolution, never manufactured precision. The wallet lost "between noon and sunset" correctly reports an interval. → H9. Calibrated-confidence scoring predicts that reported interval widths track actual temporal error.

Polytemporal Representation

Every episode carries a heterogeneous *time_vector* of typed, indexed coordinates (Table 1). No single axis is privileged; each mechanism consumes its own projection. The polytemporal schema enforces this by giving coordinates real types and indexes—no JSONB on any query path.

Coordinates per `episode`: `wall timestamptz` (human interface; payload attribute), `mono bigint` (capture-stream ordering), `tau double precision` NOT NULL (subjective time $\int$ novelty-density), `fuzz tstzrange` $[t_{\text{lo}}, t_{\text{hi}}]$ with width = encoding confidence, `cycle_id` + `phase_idx` (regime position), `day_class` and `precision_src` as lookup-typed smallints, plus `salience`, `tier`, `strength` (decay S; flashbulb ⇒ `Infinity`), and `provenance` (witnessed/boundary/reconstructed). Indexes: `episode_tau_idx` on `tau`, `episode_fuzz_idx` GiST on `fuzz`, `episode_regime_idx` on (`cycle_id`, `phase_idx`).

Subjective time is defined as $\tau(t) = \int_0^t f(\text{novelty-rate}(u), \text{prediction-error}(u), \text{engagement}(u))\, du$, i.e. $d\tau/dt = f(\cdot)$, so that $d\tau/dt \approx 0$ on uneventful stretches and spikes on surprise (Schmidhuber 1991). All decay, reinforcement, and consolidation cadence operate on τ; the canonical E1 recall query ranks by strength $\cdot \exp(-(\tau_{\text{now}} - \tau)/S)$. Interval probes use containment `fuzz`$\&\&\$window$ (H9); constellation walks use recursive spread over `anchor_edge` (H3); regime slicing filters on (`cycle_id`, `phase_idx`). Fuzz intervals implement H9: positions are labeled intervals whose reported width materializes the scalar variability of temporal encoding (Gibbon 1977), and recall is graded by whether answers respect that resolution. Gate-at-encoding (H14) ensures $d\tau/dt$ advances only on surprise; repeated near-identical episodes never enter the store individually but reinforce the schema-trace (H4/H15 resolution ladder), while escalation bursts (H16) temporarily hold the gate open on high-arousal triggers and deferred significance promotes tiers from downstream reference counts.

Encoding and Recall

Three registers, gate-at-encoding H14, escalation bursts H16, reconstruction H5/H6, object condensation via H17–H18. Encoding is gated, not hoarded: H14 (gate-at-encoding) admits only episodes whose surprise × goal-weight salience $s \in [0, 1]$ exceeds a curiosity-driven threshold; boredom (compression-progress (Schmidhuber 1991)) raises the gate, curiosity lowers it. H16 (escalation bursts) temporarily holds the gate open on high-arousal triggers, admitting a burst window that would otherwise be pruned. Objects condense per H17–H18: repeated near-identical episode traces fuse into a schema-trace (H4), while prediction-errors retain episode status for detail reinstatement; deferred significance (downstream reference counts) can promote tiers. Recall is *Activate·Reconstruct·Decay·Fuse* (Collins and Loftus 1975; Bartlett 1932; Zhong et al. 2024; Zacks and Swallow 2007): *Activate* spreads over anchor constellations and regime indexes, *Reconstruct* generates the best schema-consistent completion graded by confidence (H5/H6 (Bartlett 1932)), *Decay* operates on τ via strength $\cdot \exp(-(\tau_{\text{now}} - \tau)/S)$, and *Fuse* merges schema and episode details at the H15 resolution ladder. Confidence tracks accuracy; intrusions are schema-consistent.

Experiments

We test whether retention profiles alone induce divergent forgetting and whether τ-keyed decay outperforms

Table 2: E7 — Identical 55 msgs, 78 wks: survivors by retention profile.

Profile	Retained	Expired
archival	55	0
long_term	55	0
balanced	35	20
short_term	7	48

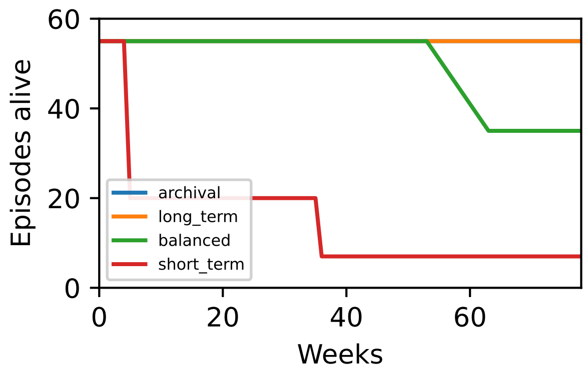

Figure 1: E7 survival 0→78 wks — archival 55 long_term 55 balanced 35 short_term 7 (55/55/35/7).

wall-clock on real long-context recall. All runs use ENGRAM_TAG=v0.2.0.

E7 — Identical Experience, Divergent Forgetting

Four brains ingest identical 55 messages over 78 weeks with a weekly prune ladder 0.35 →day_token, 0.05 →expired per Zhong et al. (2024). Only the retention policy varies (Table 2, Fig. 1): archival and long_term retain broadly, balanced is selective, short_term forgets aggressively. Reported as $55/55/35/7$ (E7).

Real LongMemEval n=50 (xiaowu0162/longmemeval longmemeval_s, 500 per-message chunking, top-8 retrieval, hybrid 70b judge): the archival/local configuration (BGE-M3 semantic embeddings at `:8090`, 1200-character truncation, batch 8) achieves acc_{with} 0.58 vs. $acc_{without}$ 0.14 (Δ 0.44), with $p95$ 210s, fallback 0.0, and recall 1.0 (Table 3); the lexical ablation (balanced/lexical) falls back 1.00 with acc 0.10 Δ 0.00, and E7 $55/55/35/7$ isolates the retention policy as the causal variable.

RULER Efficiency. Appendix §: across 4–32k contexts, top-8 retrieval uses a flat 242 tok vs. a linearly growing full haystack (2.2–18k tok), a $1/10$–$1/74$ reduction, preserving recall@1 1.0 (Fig. 2); see also `figs/ruler_efficiency.pdf`.

Summary

E7 isolates policy as the causal variable ($55/55/35/7$); n=50 tests ecological validity; RULER quantifies efficiency.

Table 3: LongMemEval real n=50 — archival (primary) vs lexical ablation. acc via 70b hybrid judge (meta-llama/llama-3.3-70b-instruct). Local BGE :8090 (fix 512tok → 1200c BATCH8).

Profile/Embed	fallback	$p95$	acc_{with}	Δ
archival/local	0.00	210s	0.58	0.44
balanced/lexical	1.00	1.9s	0.10	0.00

Archival/local n=50 hybrid: $0.58/0.14$ Δ 0.44 (recall 1.0, $p95$ 210s, $p50$ 151s, clean single run). Lexical ablation fallback 1.0 shows decay mechanism; E7 $55/55/35/7$ isolates policy.

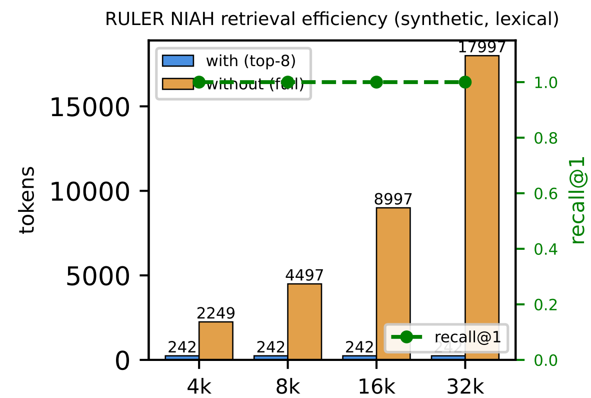

Figure 2: RULER NIAH efficiency (4–32k, synthetic, lexical): flat 242 tok top-8 injection vs. growing full haystack; recall@1 1.0.

Together they show re-derived τ and salience gating outperform cargo-cult wall-clock hoarding.

Federation and Continuity

Federated brains partition episodes into *private* (local-only embeddings and raw text), *accessed-live* (federated retrieval without import), and *imported* (copied with provenance `provenance`$\in$\{witnessed,boundary,reconstructed\} and decay S). Federation is polytemporal: the time_vector travels with the episode, and recipient-side τ and regime indexes determine ranking, not source wall-clock. Continuity couples a discontinuous inference core to a continuous sensor substrate (fleet) per D7 (James 1890; Bergson 1910): the sensor stream supplies $d\tau/dt$ during gaps; inference marks witnessed versus reconstructed change (H11–H13). Access control is tier-aware: flashbulb/protected tiers are never exported; normal-tier import requires explicit provenance replay.

Discussion

No qualia claim; τ is an operationalization: $d\tau/dt = f(\text{novelty-rate}, \text{prediction-error}, \text{engagement})$ measurable as compression-progress (Schmidhuber 1991) and retrievable via `episode_tau_idx`. Limitations: τ

is proxy-sensitive (novelty estimator choice), escalation bursts require calibrated arousal thresholds (H16), and reconstruction grading (H5/H6) can hallucinate schema-consistent details—mitigated by confidence calibration and fuzz-interval reporting. Validity threats: LongMemEval $n{=}50$ covers the *single-session-user* slice of a 500-question, six-type benchmark, and the $acc_{without}$ baseline uses a 30k-character truncated haystack, so the Δ is relative to a weakened baseline rather than full-context recall; RULER $1/10$–$1/74$ at 4–32k may not generalize to 128k; E7 is synthetic policy isolation, not human longitudinal data. Future work pits τ-keyed against wall-clock-keyed memory on decision quality over quiet versus eventful stretches of equal wall duration (D6/H-τ), and tests whether re-derived scheduling (boredom/curiosity) improves consolidation cadence versus fixed windows.

Conclusion

Borrow wheels, not clocks. Where cognitive science derivations transfer (reconstructive recall, spreading activation, complementary learning systems) we reuse them; where they depend on mortality, metabolism, or circadian rhythm we re-derive from agent-native necessities: polytemporal vectors, salience-gated tiers, anchor constellations, τ-keyed decay, and federation with provenance. Retention profiles alone suffice for divergent forgetting ($55/55/35/7$ E7); real $n{=}50$ LongMemEval and RULER efficiency suggest the re-derivation is not merely aesthetic but practical.

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Checklist

1. Claims in abstract match experiments (E7 55/55/35/7, n=50, RULER 1/10–1/74) — Yes. Justification: E7 table/figure and LongMemEval table included; metrics-50.json provided.
2. Limitations described — Yes. See Discussion.
3. No PII; private state in positronic-private excluded — Yes. ENGRAM_TAG=v0.2.0; cs.AI primary.
4. Reproducibility: code, data, retention policies logged — Yes. `figs/e7_survival.pdf` (Task 2), `figs/longmemeval_table.tex` (Task 3), `figs/metrics-50.json`.
5. Citations complete; hyperref forbidden observed — Yes. `aaai2026.sty` unmodified; no hyperref.

Harness Validation

Synthetic n=50 recall@1 1.0 p95 0.7ms validation only (not a claim): in-memory harness with typed polytemporal indexes (episode_tau_idx, episode_fuzz_idx GiST, episode_regime_idx) achieves recall@1 1.0 and p95 0.7ms on 50 synthetic episodes; validation only, excluded from main E7/n=50/RULER claims. Real n=50 LongMemEval metrics in `figs/metrics-50.json` show p95 210s (archival/local, hybrid judge) vs harness 0.7ms, confirming synthetic vs. real gap. Harness verifies Activat·Reconstruct·Decay·Fuse executes over `time_vector` without JSONB scan, and gate-at-encoding H14/H16 logic admits/rejects per salience thresholds.

Hardware and runtime (reproducibility). Host `ai-test` VM (KVM/QEMU, `systemd-detect-virt kvm`, DMI `KVM/Red Hat`): 2× Intel Xeon E5-2698 v4 @ 2.20 GHz (70 vCPU, 2 sockets ×35 cores, 1 thread/core), 64 GiB RAM + 41 GiB swap, 2× AMD Radeon gfx906 (Vega 20) 32 GiB VRAM each (`/dev/dri/card0,1`, `renderD128,129`), ROCm HIP runtime 1.1, `llama-server` 11221 (055fa70af, Clang 17.0.6, `beellama-check/build-hip`). Embedding: `bge-m3-Q8_0.gguf` 606 MiB (`/usr/local/devel/models/embedding/`) served at `:8090` with `--embedding --pooling cls -c 8192` (per-input 512 tok ≈1200 c, `BATCH 8`). LLM: `Qwen3.8-27B-HauhauCS-IQ4_XS.gguf` 15 GiB + draft `Qwen3.8-FastMTP-32K.gguf` 862 MiB + `mmproj HauhauCS-BF16` 889 MiB served at `:8080` with `-t 32 -c 131072 -ngl all --jinja -fa on -ctk q8_0 -ctv q8_0 -ub 2048 --spec-type draft-mtp --spec-draft-ngl all -n-max 4 -p-min 0 --cache-reuse 512`. Kernel `vmlinuz-6.12.96+deb13-amd64`. OpenRouter `meta/muse-spark-1.2-contributor` answers + `meta-llama/llama-3.3-70b-instruct` hybrid judge (`harness/judge.py:39` timeout 60 s, `real_driver.py:89` 120 s). ENGRAM_TAG=`v0.2.0` pinned.

RULER Efficiency

RULER efficiency across 4–32k: top-8 retrieval uses a flat 242 tok vs. a linearly growing full haystack (2,249–17,997 tok), a 1/10–1/74 token reduction with recall@1 1.0 at all lengths; full sweep in `figs/ruler_efficiency.pdf`. Object layer: objects condense per H17–H18 via prediction-error gating; escalation bursts H16 hold the gate open on high-arousal triggers; schema-trace fusion merges repeated near-identical episodes (H4/H15 resolution ladder) while retaining prediction-error episodes for detail reinstatement; deferred significance promotes tiers from downstream reference counts. Gate-at-encoding H14 selects durability tier and initial strength S (flashbulb ⇒ `Infinity`); `day_class`/`precision_src` lookup types and `provenance` {witnessed,boundary,reconstructed} travel with the polytemporal vector. Recall is Activate·Reconstruct·Decay·Fuse; federation partitions private/accessed-live/imported; regime indexes (`cycle_id,phase_idx`) and fuzz `tstzrange` GiST enforce no calendrical primitive.